

Regular Wave Solutions of the Interior Regge-Wheeler and Zerilli Equations

Inside a Thin Spherical Shell

Setup: a thin spherical shell of matter sits at areal radius $r = R$. Outside ($r > R$) the spacetime is Schwarzschild with total mass M ; inside ($r < R$) it is vacuum. We solve the master (Regge-Wheeler/Zerilli) equations of Martel & Poisson (2005) [1] for a general perturbation of this interior region, in the frequency domain, imposing regularity at $r = 0$.

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1 The interior background is Minkowski

By Birkhoff's theorem, any spherically symmetric vacuum region of spacetime is (locally) isometric to a piece of the Schwarzschild solution with some mass parameter M_{int} ,

$$ds^2 = -f_{\text{int}} dt^2 + f_{\text{int}}^{-1} dr^2 + r^2 d\Omega^2, \quad f_{\text{int}} = 1 - \frac{2M_{\text{int}}}{r}.$$

The interior region $0 \leq r < R$ is vacuum (the shell carries all the stress-energy, concentrated at $r = R$) and must be regular at $r = 0$: the Kretschmann scalar of Schwarzschild, $R_{abcd}R^{abcd} = 48M_{\text{int}}^2/r^6$, diverges at $r = 0$ unless $M_{\text{int}} = 0$. Hence

$$\boxed{f_{\text{int}} \equiv 1} \quad \implies \quad ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2 \quad (0 \leq r < R), \quad (1)$$

i.e. the interior is exactly flat Minkowski spacetime in spherical coordinates. We work with this background throughout.

2 The master equations and their $M \rightarrow 0$ limit

2.1 General- M equations (Martel & Poisson 2005 [1])

Outside sources, the even-parity (Zerilli-Moncrief) and odd-parity (Cunningham-Price-Moncrief) master functions $\Psi_{\text{even}}^{lm}(t, r)$, $\Psi_{\text{odd}}^{lm}(t, r)$ satisfy covariant wave equations on the (t, r) submanifold $\mathcal{M}^2$,

$$(\square - V_{\text{even}})\Psi_{\text{even}}^{lm} = S_{\text{even}}^{lm}, \quad (\square - V_{\text{odd}})\Psi_{\text{odd}}^{lm} = S_{\text{odd}}^{lm}, \quad (2)$$

with $\square := g^{ab}\nabla_a\nabla_b$ and potentials

$$V_{\text{even}} = \frac{1}{\Lambda^2} \left[\mu^2 \left(\frac{\mu+2}{r^2} + \frac{6M}{r^3} \right) + \frac{36M^2}{r^4} \left(\mu + \frac{2M}{r} \right) \right], \quad (3)$$

$$V_{\text{odd}} = \frac{l(l+1)}{r^2} - \frac{6M}{r^3}, \quad (4)$$

where $\mu := (l-1)(l+2)$ and $\Lambda := \mu + 6M/r$. In tortoise coordinates ($dr_*/dr := f^{-1}$, $f = 1 - 2M/r$) one has $\square\Psi = f^{-1}(-\partial_t^2\Psi + \partial_{r_*}^2\Psi)$, so these equations take the more familiar scattering form

$$-\partial_t^2\Psi_P + \partial_{r_*}^2\Psi_P - \bar{V}_P\Psi_P = fS_P, \quad \bar{V}_P := fV_P, \quad P \in \{\text{even}, \text{odd}\}. \quad (5)$$

2.2 Interior limit $M = 0$

Inside the shell there is no matter for $0 < r < R$ (the shell's stress-energy is a distribution supported at $r = R$ only), so $S_{\text{even}} = S_{\text{odd}} = 0$ there. Setting $M = 0$:

$$\Lambda|_{M=0} = \mu, \quad V_{\text{even}}|_{M=0} = \frac{1}{\mu^2} \left[\mu^2 \frac{\mu+2}{r^2} \right] = \frac{\mu+2}{r^2} = \frac{l(l+1)}{r^2}, \quad (6)$$

using $\mu+2 = (l-1)(l+2) + 2 = l(l+1)$. Likewise

$$V_{\text{odd}}|_{M=0} = \frac{l(l+1)}{r^2}. \quad (7)$$

So the even- and odd-parity potentials coincide inside the shell: $V_{\text{even}} = V_{\text{odd}} = l(l+1)/r^2$. This is expected — flat spacetime alone cannot distinguish the two parities dynamically at the level of the master scalar; the distinction re-enters only through how each Ψ_P is built from the metric perturbation, and through the (different) junction conditions each imposes at $r = R$.

Since $f \equiv 1$, the tortoise coordinate is simply $r_* = r$, $\bar{V}_P = V_P$, and $\square\Psi = -\partial_t^2\Psi + \partial_r^2\Psi$. Both master functions therefore obey the *same* source-free wave equation inside the shell:

$$\boxed{\left(-\frac{\partial^2}{\partial t^2} + \frac{\partial^2}{\partial r^2} - \frac{l(l+1)}{r^2} \right) \Psi_{lm}^P(t, r) = 0}, \quad P = \text{even, odd}, \quad 0 < r < R. \quad (8)$$

3 Frequency-domain solution: the Riccati-Bessel equation

3.1 Separation of variables

Write a monochromatic mode as $\Psi_{lm}^P(t, r) = e^{-i\omega t}\psi_l(r)$ (the radial equation does not depend on P , m , or the sign of ω). Substituting into (8),

$$\psi_l''(r) + \left(\omega^2 - \frac{l(l+1)}{r^2} \right) \psi_l(r) = 0. \quad (9)$$

This is the *Riccati-Bessel equation* [2, 3] (see Appendix A for its general form and derivation). With $x := \omega r$, its two independent solutions are

$$\hat{j}_l(x) := x j_l(x), \quad \hat{y}_l(x) := x y_l(x), \quad (10)$$

where j_l, y_l are the ordinary spherical Bessel functions of the first and second kind (solutions of $x^2 u'' + 2xu' + [x^2 - l(l+1)]u = 0$).

3.2 Behavior at $r = 0$

As $x \rightarrow 0$,

$$j_l(x) \rightarrow \frac{x^l}{(2l+1)!!}, \quad y_l(x) \rightarrow -\frac{(2l-1)!!}{x^{l+1}}, \quad (11)$$

so that

$$\hat{j}_l(x) \rightarrow \frac{x^{l+1}}{(2l+1)!!} \text{ (regular, vanishes as } r^{l+1}\text{),} \quad \hat{y}_l(x) \rightarrow -\frac{(2l-1)!!}{x^l} \text{ (singular for } l \geq 1\text{).} \quad (12)$$

Regularity of the perturbed metric at the center of symmetry requires the standard multipole falloff $\psi_l(r) \sim r^{l+1}$ as $r \rightarrow 0$ (the same condition that selects j_l over y_l for any regular source-free multipole field, e.g. electromagnetic or scalar, in flat space); the $\hat{y}_l$ branch must be discarded even for $l = 0$, where it stays finite ($\hat{y}_0(x) = -\cos x$) but does not vanish, hence does not correspond to a smooth (Cartesian-analytic) field at the origin. We therefore keep only

$$\psi_l(r) \propto \hat{j}_l(\omega r) = \omega r j_l(\omega r). \quad (13)$$

3.3 Behavior at large argument — these are genuine waves

As $x \rightarrow \infty$,

$$\hat{j}_l(x) \rightarrow \sin\left(x - \frac{l\pi}{2}\right), \quad (14)$$

i.e. an oscillatory, bounded (not exponentially growing or decaying) function of r for real ω — a standing spherical wave, regular at the center, exactly as required.

3.4 Explicit low multipoles

Using $\hat{j}_0(x) = \sin x$ and the recursion $\hat{j}_{l+1}(x) = \frac{2l+1}{x} \hat{j}_l(x) - \hat{j}_{l-1}(x)$ (with $\hat{j}_{-1}(x) := \cos x$):

$$\hat{j}_0(x) = \sin x, \quad (15)$$

$$\hat{j}_1(x) = \frac{\sin x}{x} - \cos x, \quad (16)$$

$$\hat{j}_2(x) = \left(\frac{3}{x^2} - 1\right) \sin x - \frac{3}{x} \cos x, \quad (17)$$

$$\hat{j}_3(x) = \left(\frac{15}{x^3} - \frac{6}{x}\right) \sin x - \left(\frac{15}{x^2} - 1\right) \cos x. \quad (18)$$

(Only $l \geq 2$ carries gravitational-wave degrees of freedom; $l = 0, 1$ are non-radiative, as established in the companion formulario. $\hat{j}_2$ is thus the physically leading case.)

4 Result: the regular interior solution

4.1 Single mode

For a single (real) frequency ω , the regular, oscillatory (wave) solution of both the interior Zerilli and interior Regge-Wheeler equations is

$$\Psi_{lm}^{\text{even}}(t, r) = A_{lm}^{\text{even}}(\omega) \hat{j}_l(\omega r) e^{-i\omega t}, \quad \Psi_{lm}^{\text{odd}}(t, r) = A_{lm}^{\text{odd}}(\omega) \hat{j}_l(\omega r) e^{-i\omega t}, \quad (19)$$

with $A_{lm}^{\text{even}}(\omega), A_{lm}^{\text{odd}}(\omega) \in \mathbb{C}$ arbitrary complex constants (amplitude and phase combined) for each (l, m, ω) — the two parities share the same radial profile $\hat{j}_l(\omega r)$ but carry independent, in general complex, amplitudes, since they are independent physical degrees of freedom.

4.2 General regular wave packet

By linearity, the most general regular solution of (8) is a superposition over frequency, i.e. a Fourier integral in t alone:

$$\boxed{\Psi_{lm}^P(t, r) = \int_{-\infty}^{\infty} d\omega A_{lm}^P(\omega) \hat{j}_l(\omega r) e^{-i\omega t}, \quad P = \text{even, odd},} \quad (20)$$

with $A_{lm}^P(\omega)$ an arbitrary complex amplitude density, fixed in a full problem by initial data or by matching to the exterior Schwarzschild solution across the shell at $r = R$ (not treated here). ($\hat{j}_l(\omega r)$ is well defined for $\omega < 0$ too, since $j_l(-x) = (-1)^l j_l(x)$ for integer l ; the equation only ever depends on ω^2 .) Every term is finite at $r = 0$; the real and imaginary parts are each purely oscillatory in t and (asymptotically) in r — a regular standing gravitational wave confined to the flat interior.

A The Riccati-Bessel equation: general form

Bessel's equation of order ν (real, not necessarily integer),

$$x^2 \frac{d^2 Z_\nu}{dx^2} + x \frac{dZ_\nu}{dx} + (x^2 - \nu^2) Z_\nu = 0, \quad (21)$$

is solved by the ordinary Bessel functions $J_\nu(x), Y_\nu(x)$ (or the Hankel functions $H_\nu^{(1,2)}(x)$). The *spherical* Bessel functions of order l are defined by

$$u_l(x) := \sqrt{\frac{\pi}{2x}} Z_{l+1/2}(x), \quad (22)$$

and satisfy the spherical Bessel equation

$$x^2 u_l''(x) + 2x u_l'(x) + [x^2 - l(l+1)] u_l(x) = 0. \quad (23)$$

The *Riccati-Bessel substitution* $w_l(x) := x u_l(x)$ removes the first-derivative term from (23) identically: writing $u_l = w_l/x$ and substituting,

$$x^2 u_l'' + 2x u_l' = x^2 \left(\frac{w_l''}{x} - \frac{2w_l'}{x^2} + \frac{2w_l}{x^3} \right) + 2x \left(\frac{w_l'}{x} - \frac{w_l}{x^2} \right) = x w_l'', \quad (24)$$

so that (23) becomes, after multiplying through by x , the **Riccati-Bessel equation**

$$\boxed{x^2 w_l''(x) + [x^2 - l(l+1)] w_l(x) = 0} \iff \boxed{w_l''(x) + \left[1 - \frac{l(l+1)}{x^2} \right] w_l(x) = 0.} \quad (25)$$

The two forms are equivalent (divide/multiply by x^2); the second is the one used in the body of this note, with $x = \omega r$ and $l(l+1) \rightarrow l(l+1)$ unchanged (it is already the flat-space centrifugal term). More generally, for any real order ν , replacing $l(l+1) \rightarrow \nu(\nu+1)$ in (25) gives an equation solved by $w_\nu(x) = \sqrt{\pi x/2} \mathcal{Z}_{\nu+1/2}(x)$ for any solution $\mathcal{Z}_{\nu+1/2}$ of Bessel's equation of order $\nu+1/2$; this is precisely the family of one-dimensional Schrödinger-like equations with an inverse-square (“centrifugal”) potential.

For integer l , a standard basis of solutions of (25) is

$$\hat{j}_l(x) := x j_l(x) = \sqrt{\frac{\pi x}{2}} J_{l+1/2}(x), \quad \hat{y}_l(x) := x y_l(x) = \sqrt{\frac{\pi x}{2}} Y_{l+1/2}(x), \quad (26)$$

with $\hat{j}_l$ regular ($\sim x^{l+1}$) and $\hat{y}_l$ singular ($\sim x^{-l}$) at $x = 0$; the combinations $\hat{h}_l^{(1,2)}(x) := \hat{j}_l(x) \pm i \hat{y}_l(x)$ (Riccati-Hankel functions) furnish the natural basis of purely outgoing/ingoing waves at $x \rightarrow \infty$, used e.g. for the *exterior* scattering problem (not needed here, since both are singular at the origin). The general solution of (25) is $w_l(x) = A \hat{j}_l(x) + B \hat{y}_l(x)$ for arbitrary constants A, B ; regularity at $x = 0$ sets $B = 0$, which is exactly the condition used in Sec. 3.2.

Reference: Abramowitz & Stegun [2], §10.3 (“Riccati-Bessel Functions,” p. 445); NIST Digital Library of Mathematical Functions [3], Chapter 10.

References

- [1] K. Martel and E. Poisson, “Gravitational perturbations of the Schwarzschild spacetime: A practical covariant and gauge-invariant formalism,” *Phys. Rev. D* **71**, 104003 (2005).
- [2] M. Abramowitz and I. A. Stegun, eds., *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*, 9th printing (Dover, New York, 1972), §10.3, p. 445.
- [3] NIST Digital Library of Mathematical Functions, <https://dlmf.nist.gov/10>, Chapter 10 “Bessel Functions.”